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PROJECT AND TEAM INFORMATION

Project Title

AI NAVIGATOR: AI Feature Discovery, Recommendation and Prompt Guidance System

Team Name: Tech Hunter	*Avengers*
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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

AI Navigator is a recommendation and knowledge-management system that helps users discover AI tools, understand their features, select a suitable tool for a task and create effective prompts. Users enter a goal such as researching a topic, analyzing data, writing content, coding, creating images, or studying. The system classifies the task, searches stored AI-tool and feature records, matches required capabilities, ranks suitable options and displays recommendations with feature-specific guidance and a task-oriented prompt. It is designed as a navigation layer rather than another chatbot. The project demonstrates C++ object-oriented programming and data structures, including classes, vectors, graph relationships, adjacency lists, searching, sorting, priority-based ranking and file handling.

Updated Project Approach and Architecture

The system follows a modular, console-based C++ design. AI tools, features, tasks, prompts and recommendations are represented using classes or structures. Each AI tool can contain multiple feature records. Capabilities are connected to tools and features through directed relationships stored as an adjacency list.

The processing pipeline is:
(1) Load or enter AI-tool, feature and capability data
(2) Accept and classify the user's task
(3) Identify required capabilities
(4) Search matching feature records
(5) Calculate match scores
(6) Rank candidates using sorting or a priority queue
(7) Show the recommended AI and feature
(8) Display a usage guide and suggested prompt
(9) Generate and save a recommendation report.

DFS/BFS can be used to traverse relationships, while vectors, hashing and file handling support data storage and retrieval. The scoring model may weight task/capability match (40%), feature match (25%), project rating (15%), ease of use (10%) and cost suitability (10%). The report specifies C++ and DSA components; a particular network protocol or external API is not required by the described core workflow.

Tasks Completed

Task Completed	Team Member
Phase-II requirements and module planning	Gaurav Bansal – confirm allocation
AI-tool, feature and task data specification	Ketan Jain – confirm allocation
Graph relationships and recommendation design	Shubhangi Garg – confirm allocation
Prompt/report design and documentation	Saanvi Jain – confirm allocation

Challenges/Roadblocks

Building a dynamic system comes with a few moving parts, but the team has a clear strategy to address each challenge. We know details change, so users can easily update their records anytime. When it comes to recommendations, we're keeping things fair and practical—using standard scoring rules and giving plain-English explanations for why a specific AI tool fits your needs. Behind the scenes, every record gets a unique ID with built-in checks to keep our data linked and error-free. We've also designed our file handling to be resilient, so missing or messy data won't throw a wrench in the system. Bringing all these pieces together takes careful coordination, but we have a clear road ssmap and are ready to execute.
To overcome these challenges, the team will develop the project step by step, test each module separately, use sample datasets, and perform complete system testing to ensure that searching, ranking, prompt generation and report creation work properly.

Tasks Pending

Implement AI-tool and feature management modules	To be assigned
Implement task input, capability matching and recommendation scoring	To be assigned
Implement graph/adjacency list, searching, sorting and optional DFS/BFS	To be assigned
Implement prompt guidance, comparison, report generation and file save/load	To be assigned
Run tests, fix integration issues and prepare evaluator demonstration	All members / confirm

Project Outcome/Deliverables

The intended deliverable is a working C++ AI Navigator console application that stores AI-tool and feature records, accepts a user's task, searches capability information, ranks suitable AI tools/features, provides feature-use guidance, and displays a task-specific prompt. Additional deliverables include AI-tool comparison, a directed graph represented with an adjacency list, recommendation scoring and ranking, file-based persistence and a formatted recommendation report. The project will include a sample dataset with at least five AI tools and multiple features, plus test cases and an evaluator-ready demonstration. Optional extensions include DFS/BFS traversal, hashing, priority-queue ranking and Tri-based feature auto complete.

Progress Overview

The Phase-II design and planning specification is documented, including the data model, modules, recommendation workflow, scoring approach, class design, file structure and expected demonstration. The attached specification does not establish what proportion of the C++ application has already been implemented, so the implementation percentage and schedule status must be filled in after checking the current repository. The next priority is to complete the core workflow: AI/feature records, task entry, capability matching, ranking and prompt output. Graph traversal, hashing and Tri auto complete are recommended enhancements and can follow once the basic recommendation path is reliable. The team should track each module as completed, in progress or pending and validate integration before the final demonstration.

Codebase Information

Language: C/C++
Repository URL: To be added
Main branch: To be confirmed
Important commits: To be listed after the implementation is pushed to the repository.
Suggested source modules: AI management, feature management, graph management, task management, recommendation engine, prompt management, comparison, file handling and report generation.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
AI-Tool and Feature Record Validation	Pending	Check IDs, required fields and duplicate records.
Task Classification and Feature Search	Pending	Try research, coding and data-analysis tasks.
Recommendation Scoring and Ranking	Pending	Confirm scores and ordering using sample records.
Graph/Adjacency-list Relationships	Pending	Verify tool-feature-capability connections.
File Save and Report Generation	Pending	Check persistence and generated output.
End-to-End Console Workflow	Pending	Validate the complete user journey.

Deliverables Progress

Phase-II specification and architecture: Documented.
AI-tool/feature sample dataset and data structures: Specified; implementation to be confirmed.
Task input, Capability matching and Recommendation ranking: Validation Pending.
Prompt guidance and AI comparison: Requirements specified, Implementation to be confirmed.
Graph traversal, File persistence and Report generation: Specified implementation pending confirmation.
Final testing and Evaluator demonstration: Pending.